

Unlocking the Potential of MBSE: From Requirements to Mission Engineering, Cybersecurity, and Supply Chain Management

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Source: <https://sparxsystems.com/mbse/>

Outline

- Why MBSE?
- MBSE Applications
- MBSE Adoption and Integration
- Industry-Focused MBSE Curriculum



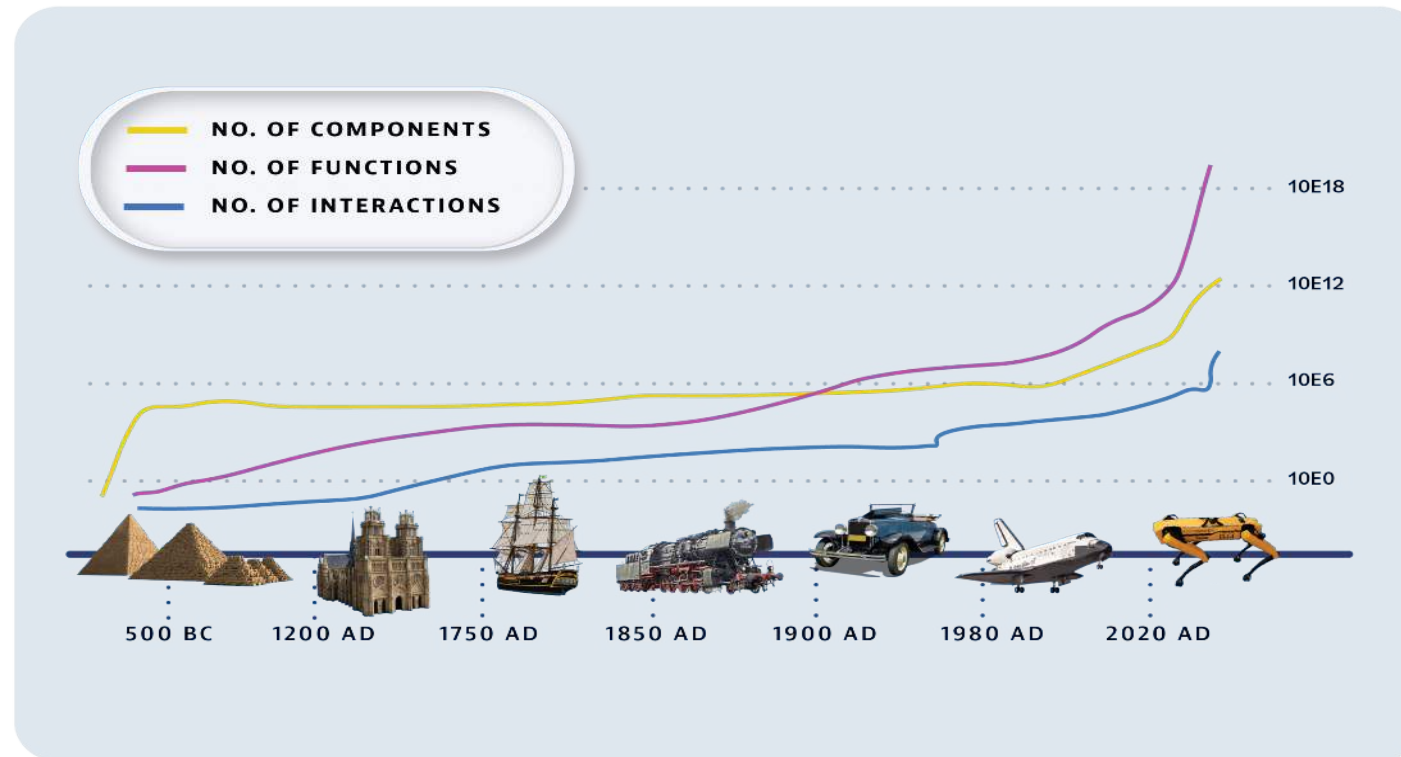
Part I: Why MBSE?



"I think you might need a few more tools"

Why Things Go Wrong – Root Causes of Problems

- Complexity

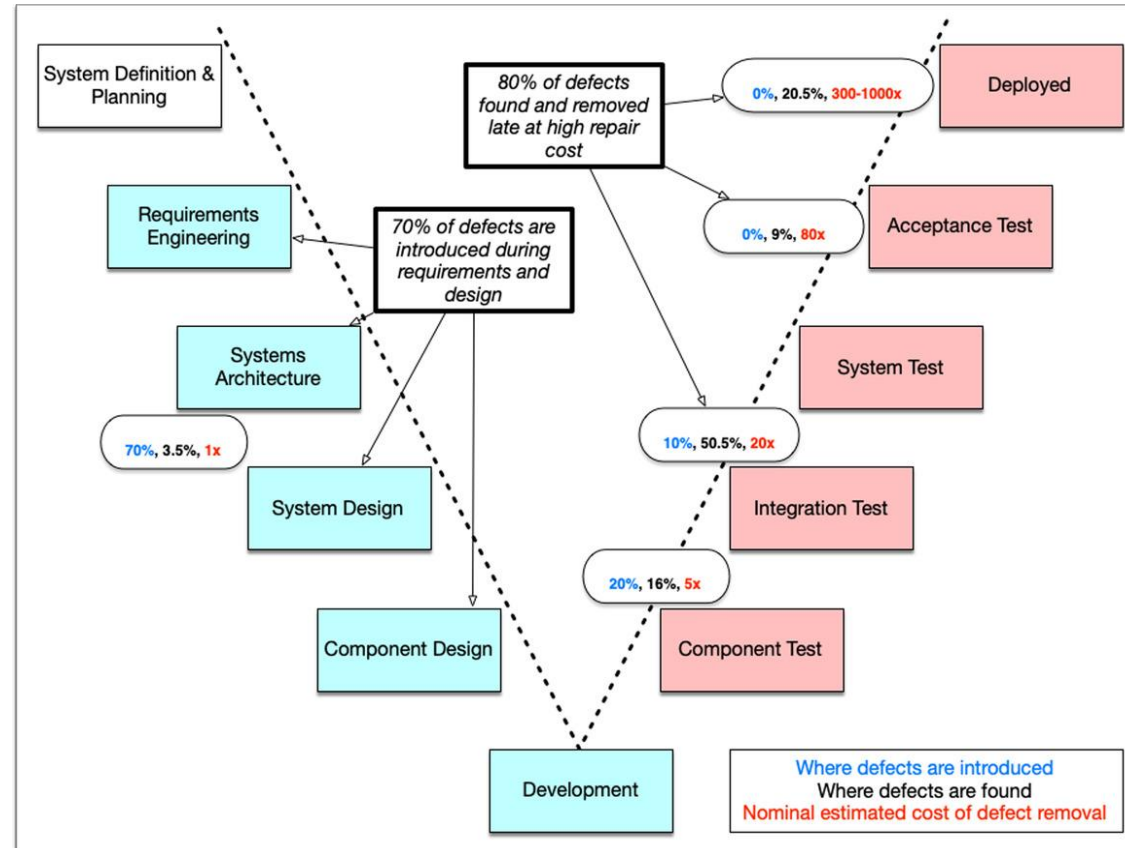


Growing complexity of systems over the year.
Source: INCOSE Systems Engineering Vision
2035

Why Things Go Wrong – Root Causes of Problems

- Communication & collaboration
 - Different stakeholders, different languages
 - Geographically, functionally, and culturally distributed workforce
- Lack of holistic lifecycle view and systems thinking
 - Life cycle mis-handling
 - Systems view (thinking) effect – suboptimal decisions – the whole is more than the sum of its parts
- Compressed development timelines
 - Driven by greater pace of change, user expectations, and evolving environments
- Security
 - Dependence on networked systems increase systems vulnerability
- Problem detection delays

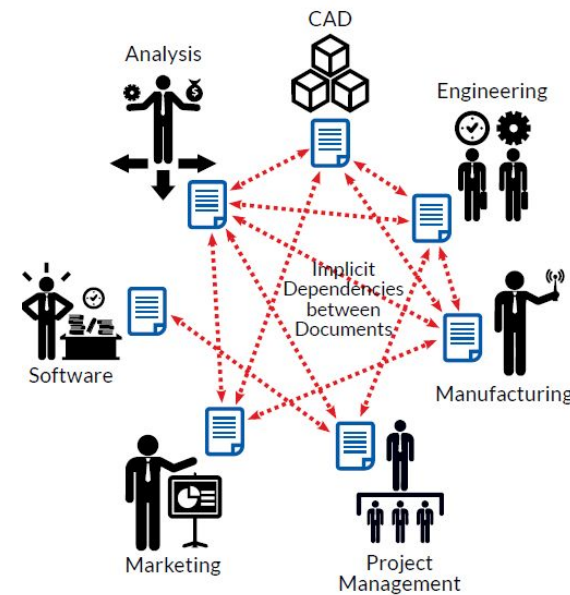
Why Things Go Wrong – Root Causes of Problems



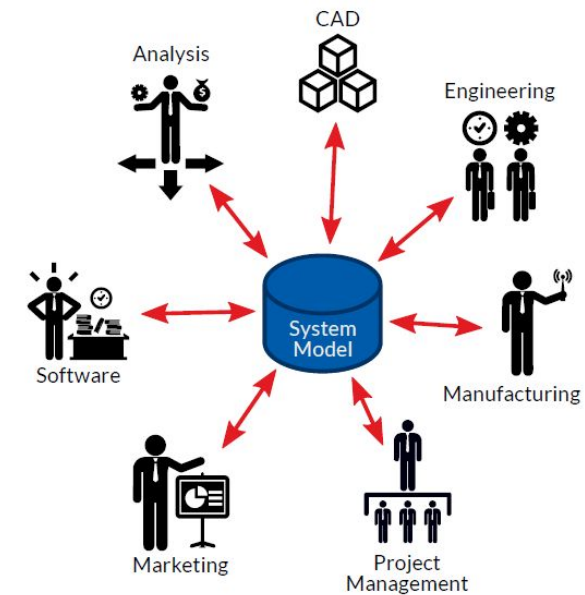
Late defect discovery escalating repair cost (Brown, 2011)

Why?

- How is the need for SE is different than the need for MBSE?
 - It isn't
- MBSE is SE in a more:
 - Rigorous
 - Repeatable
 - Demonstrable



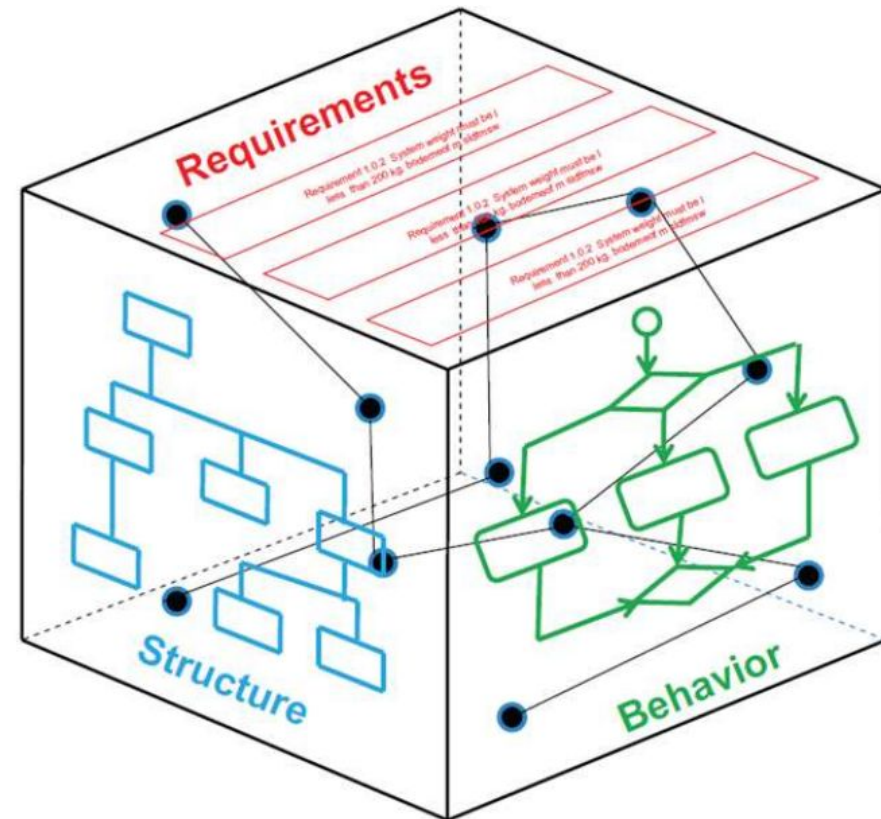
(a) Traditional document-centric SE approach



(b) Model-based SE approach

What is MBSE?

- "MBSE is the formalized application of modeling to support system requirements, design, analysis, verification, and validation activities beginning in the conceptual design phase and continuing throughout development and later lifecycle phases" (Friedenthal et al., 2007).
- SE activities are performed using the model instead of documents.
- The system model can be viewed in diagrams or in other combinations of graphical, tabular, and textual reports that are generated by querying the model and presenting the information in the desired form.
- MBSE is an approach that focuses on the digitalization of the systems engineering processes, where traditional document-based artifacts are replaced by interconnected digital models.



Richardson (2020)



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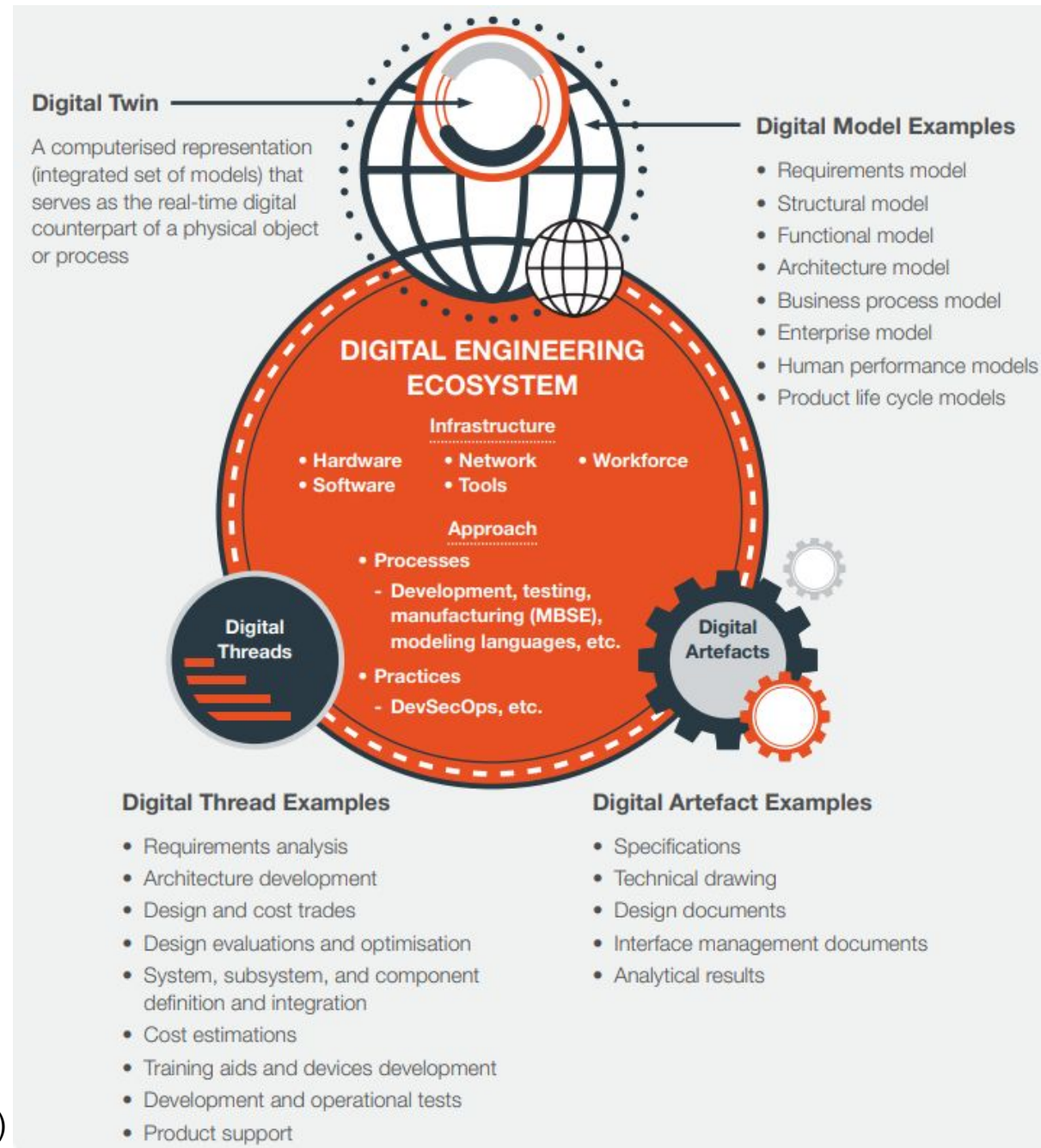
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Why Modeling?

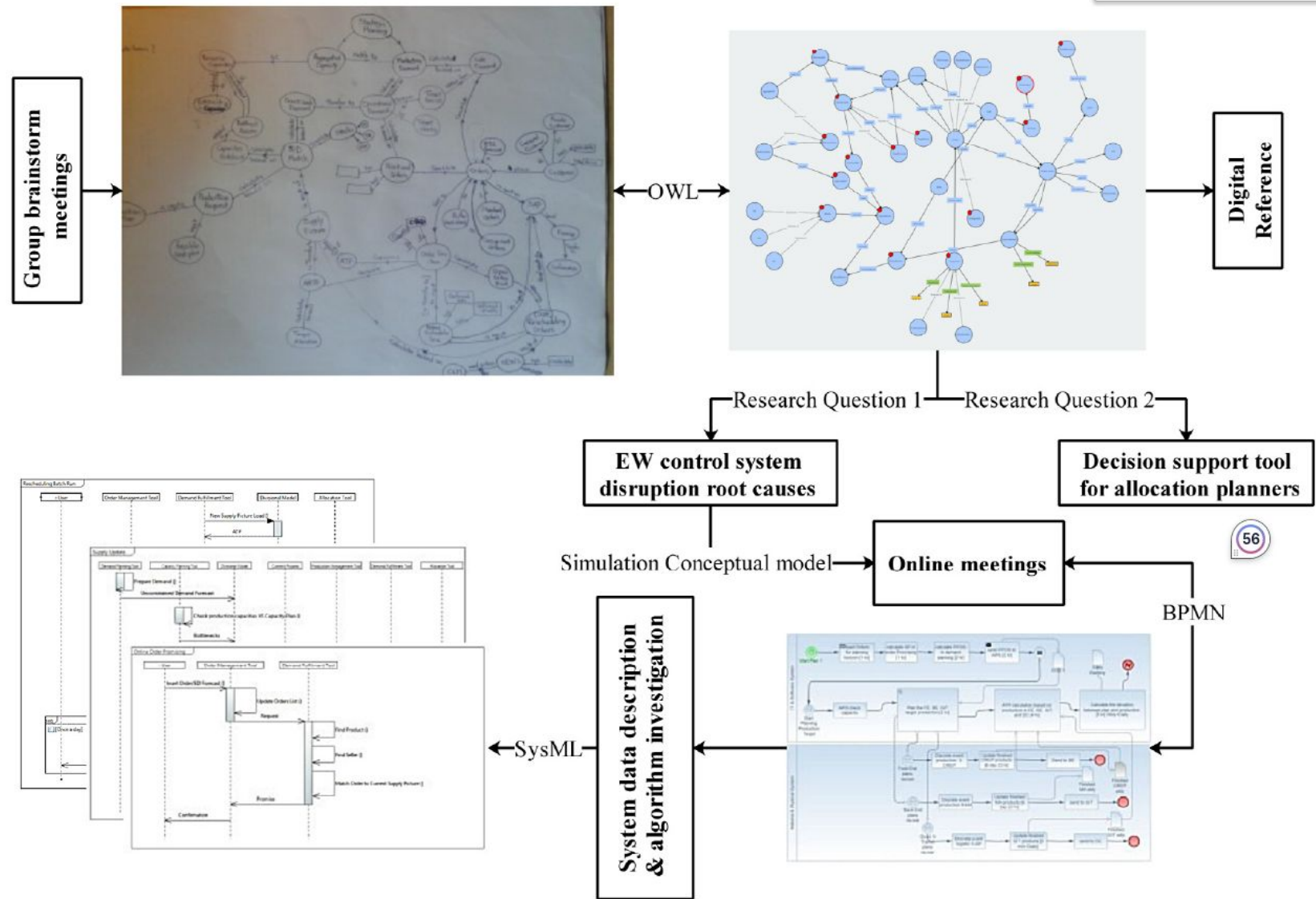
- Better understanding for design change impacts. IF you have a model, then you can do simulations!
- Easier analysis of system design before building it, e.g., integrate cybersecurity in system design, or analyze system resilience.
- Automation, Automation, Automation!
- Better consistency assurance and validation.
- Better requirements traceability.
- Facilitate iterative, and parallel system design and development.
- Knowledge preservation.



MBSE is a key component of the Digital Engineering Ecosystem

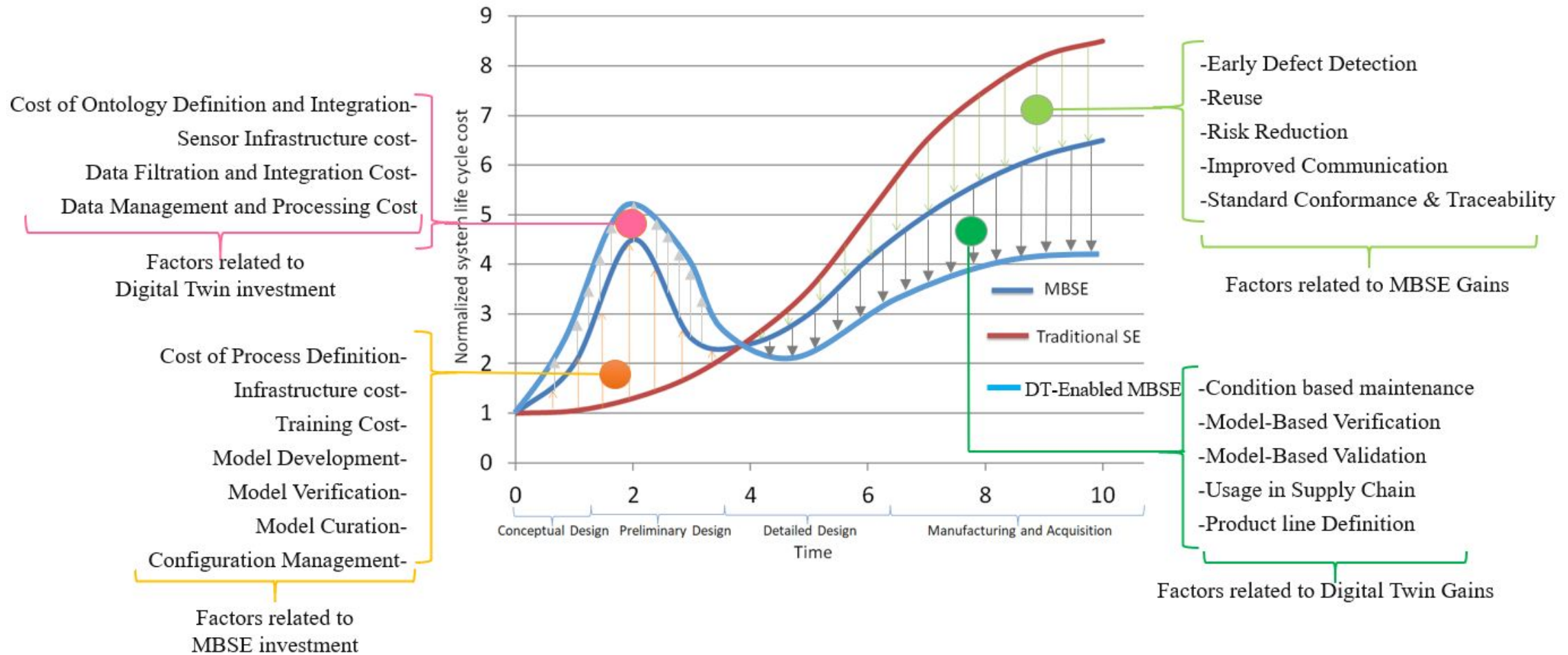


MBSE is a key stepping stone alongside two other skills to AI expert systems (i.e. OWL and BPMN) being seamlessly integrated



Pathway for implementing and using MBSE approaches for system understanding, communication with stakeholder, define research question, conceptual modeling, and depth module investigation (Mousavi et al., 2022)

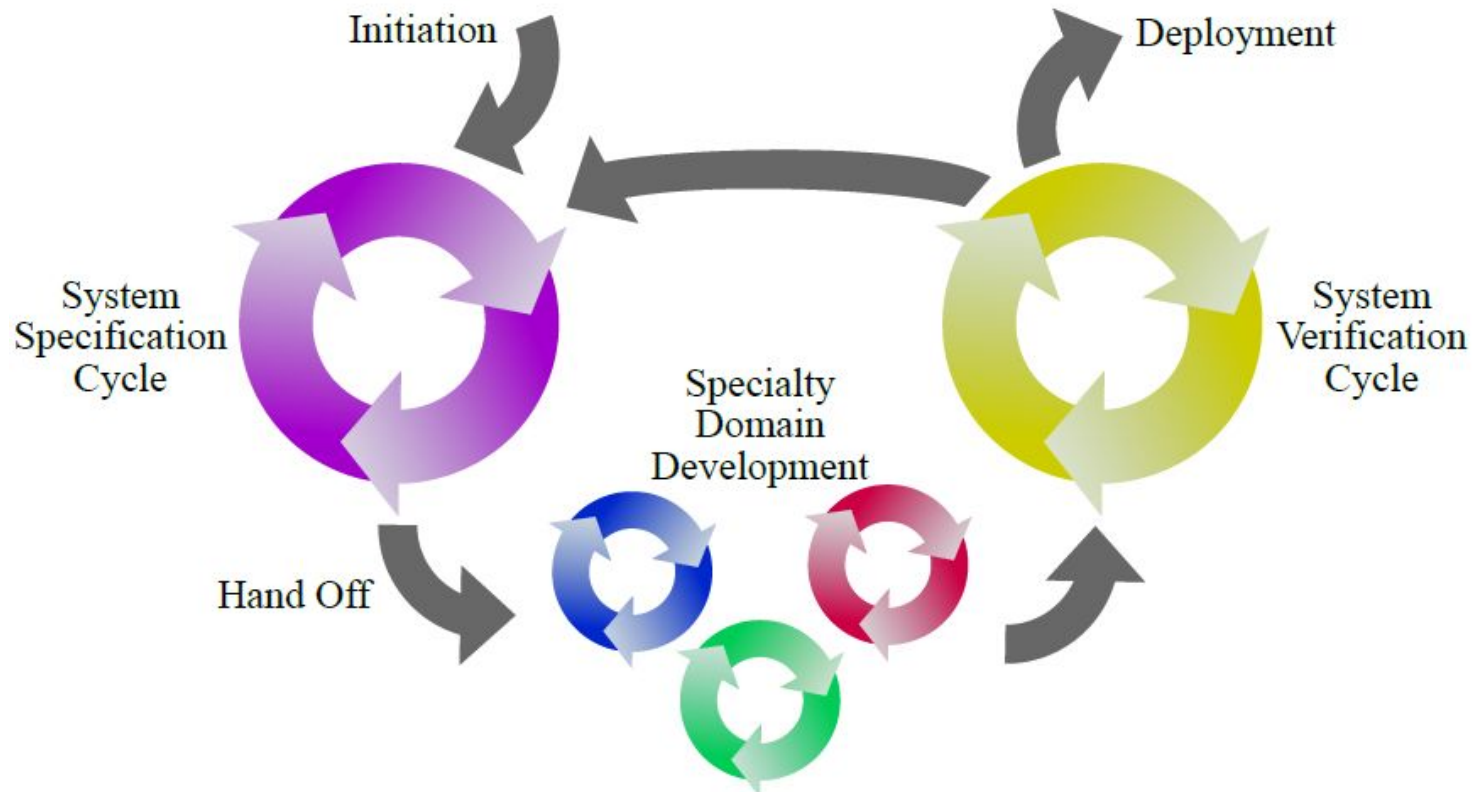
MBSE value: Economics



Comparison of costs between traditional SE and MBSE, and Digital Twin Enabled-MBSE
Madni and Purohit (2021)

Agile SE

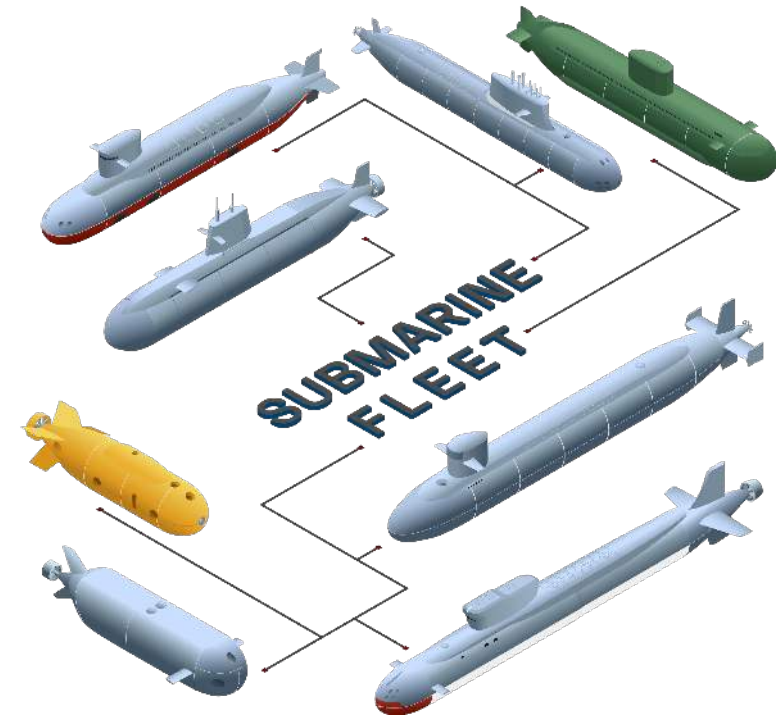
MBSE enables ‘**agile systems engineering (ASE)**’ that is a systems engineering approach that incorporates agile principles that is used to engineer systems that themselves may or may not be agile.



An Agile Systems Engineering Approach (Call and Herber, 2023)

MBSE Value: Case Study

- SWFTS (Submarine Warfare Federated Tactical Systems), a combat system family (41 sub-systems) deployed across the Royal Australian and the US navy submarine fleets supporting six submarine classes.
- Characteristics:
 - Heterogenous: sub-systems developed autonomously by different vendors.
 - Different sub-systems requirements and architectures.
 - Loosely coupled sub-systems through controlled interfaces.
 - The hosting Submarines have different sensors, weapons, and mission capabilities and the system varies based on the submarine attributes.
- Baselines need to reflect software changes to match hardware changes and introduction of new capabilities following a 2-year development, manufacturing, integration, and certification cycle.



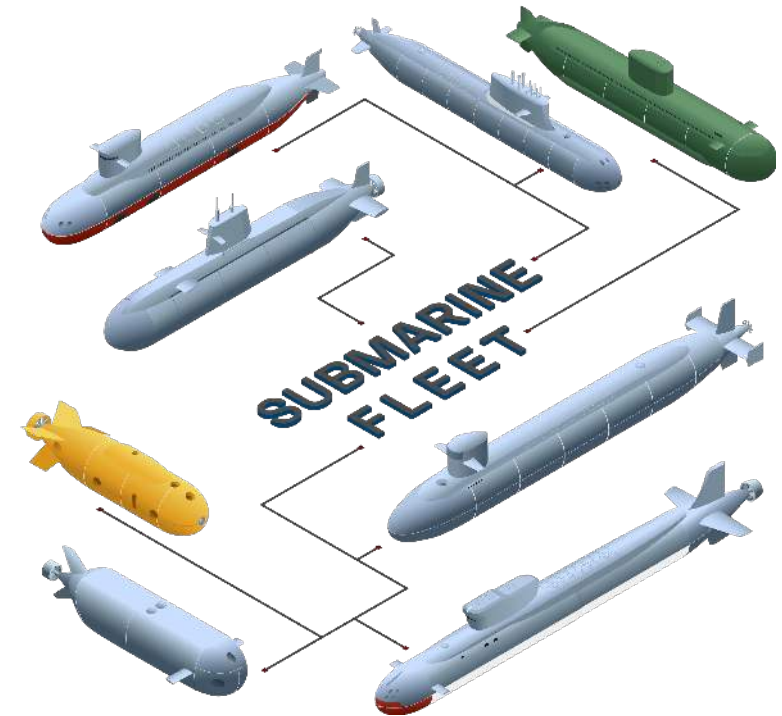
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MBSE Value: Case Study

- SE activities was performed following a document centric approach relying on conventional SE artifacts and tools, such as: UML, text files, spreadsheets, and diagrams in MS PowerPoint.
- The 3-year transition to MBSE resulted in a model that contain:
 - ~ 3,500 interface requirements
 - ~ 3,700 elements representing interfaces, methods, and data structures
 - ~ 15,000 relationships between model elements
- Recorded ROI:
 - Average number of baselines produced monthly: 30%
 - Number of integrated subsystems and combat systems variants: 60%
 - Number of discrete SE products produced by the team (directly generated from the MBSE model): 60%



Digitisation and Digitalisation



Digitisation

“converting the physical representation of information from an analogous one to a digital one in bits and bytes” (Teubner and Stockhinger, (2020)).

So what ...

Digitisation can be functionally mapped and tested; however, *digitalisation* requires *usability testing* and wherewithal for humans to exploit the capabilities, including malicious use as a cyber threat. Therefore, usability testing around *minimum viable product (MVP)* is critical to align requirements and expectations while building the wherewithal for socio-technic transformation. These principles apply not just at a whole business-level but to select user groups at an application-level.

Digitalisation

“the socio-technical change invoked by digital technologies and their confluence in digital infrastructures” (Teubner and Stockhinger, (2020))

“digital convergence, ubiquity, and connectivity in concert with the broad adoption and usage of digital technologies in multiple social and institutional contexts” (Teubner and Stockhinger, (2020))

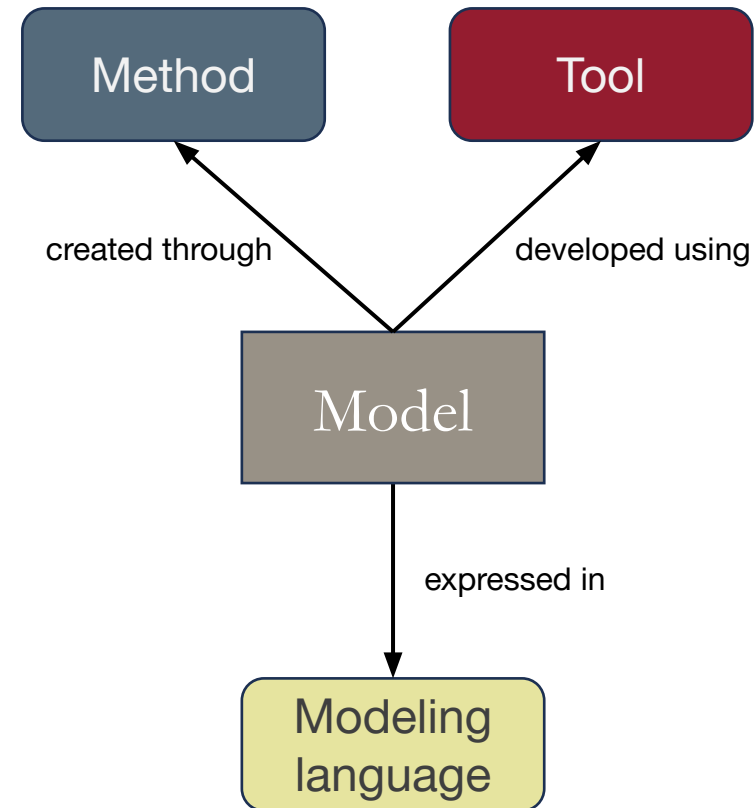
Elements of MBSE: Tools

Formal **language** used to express information or knowledge within a specific domain. It provides a set of notations and rules for constructing models that represent systems.

An MBSE **method** is the collection of related processes and methods to support the discipline of SE in a model-based or model-driven context.

A **tool** is an instrument that can enhance the efficiency of the task, if it is applied properly. The purpose of the tool is to facilitate the accomplishment of the “How”.

Example widely used in defence: No Magic Cameo Systems Modeler



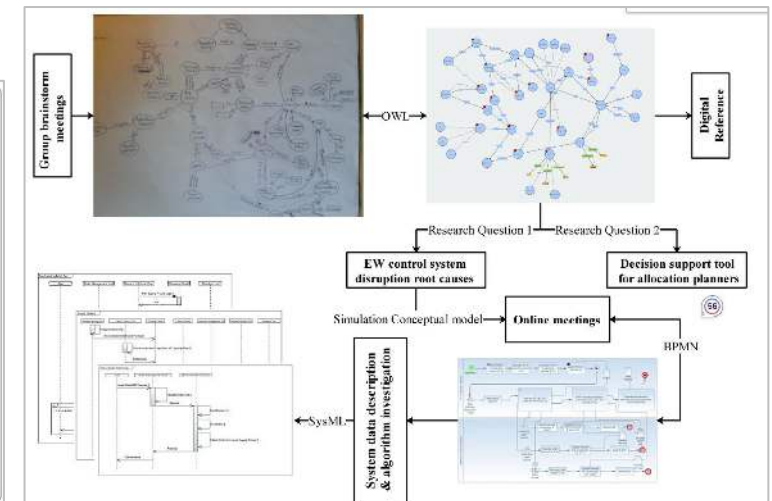
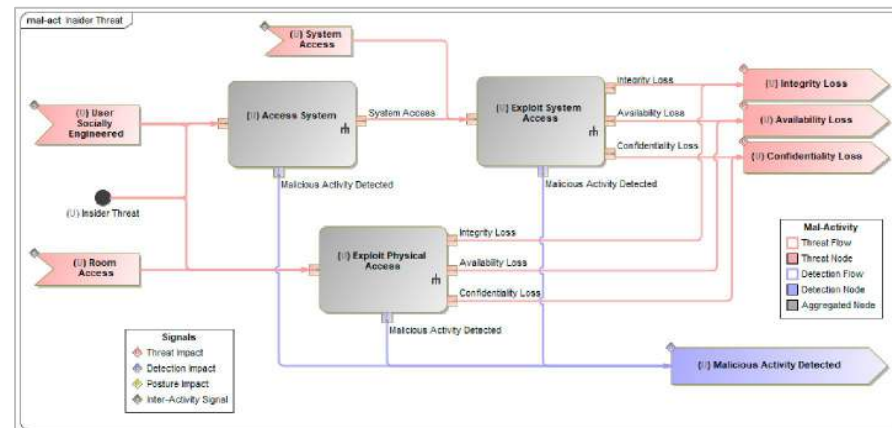
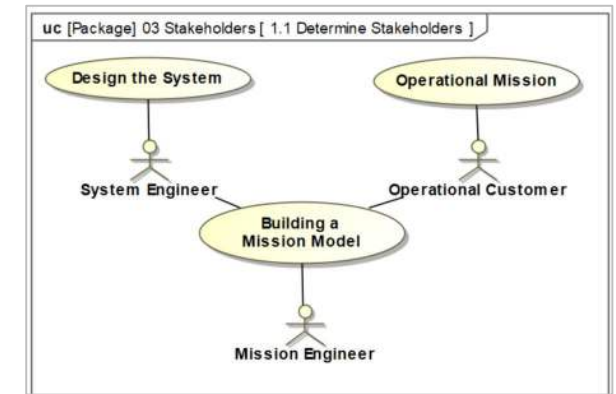
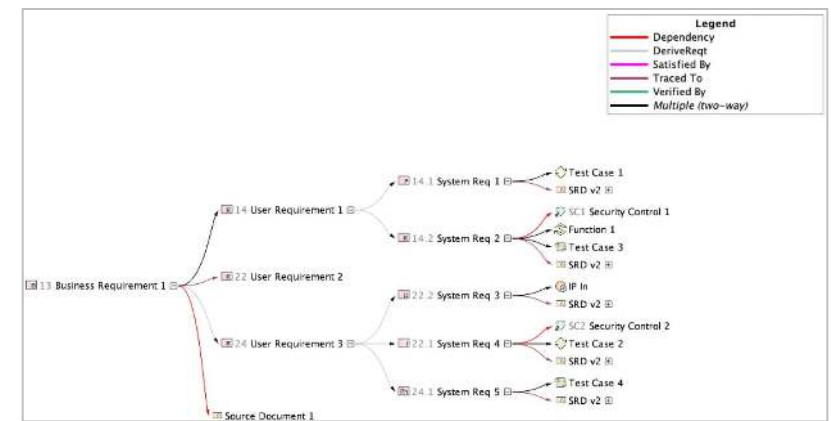
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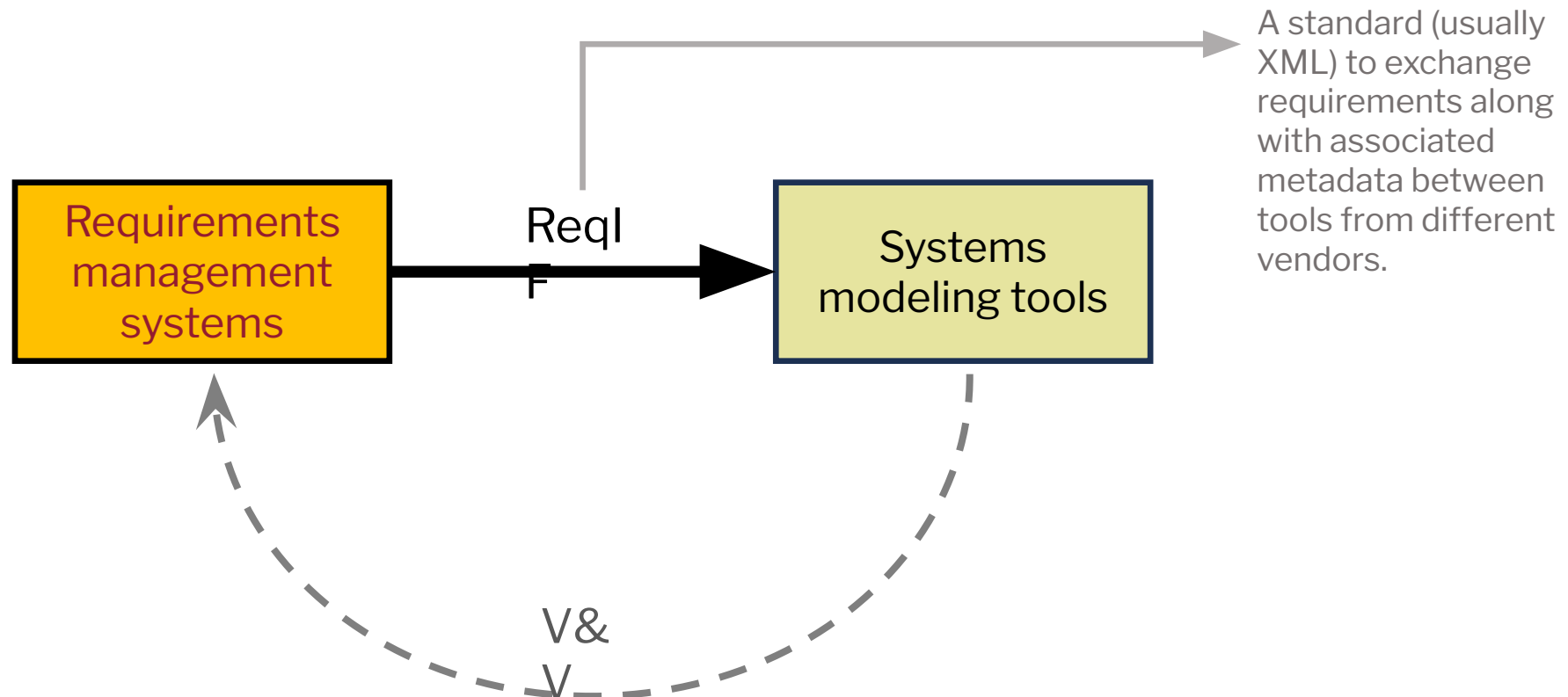
Part II: Applications of MBSE

- Robust requirements management
- Mission engineering
- Criticality of cybersecurity assessments
- Supply chain management



Robust Requirements Management

Models are not intended to replace textual requirements; **model elements and textual requirements complete each other** and must typically be related to each other, model elements help to formalize a textual requirement, for example, stakeholder requirements.



Robust Requirements Management

There are benefits (and sometimes requirements) to being able to demonstrate **traceability** from textual requirements to the system behaviors that satisfy those requirements and the structural elements to which those behaviors are allocated. With this traceability, stakeholders can be assured that all requirements are accounted for in the design and allow for system design trade space exploration before development costs are incurred (Call and Herber, 2023).

An MBSE approach affords this traceability in a way that is prohibitively difficult in a document-based approach. Using any number of commercially available modeling tools and a robust modeling methodology, the process of creating, maintaining, and reporting on these traceability links can be somewhat natural and eventually trivial.

Traceability is:

- a key advantage sought, particularly realising benefits in testing, validation, and through-life adaptability that offset its upfront investment.
- a key validation of the MBSE model itself and a KPI for DE implementation.



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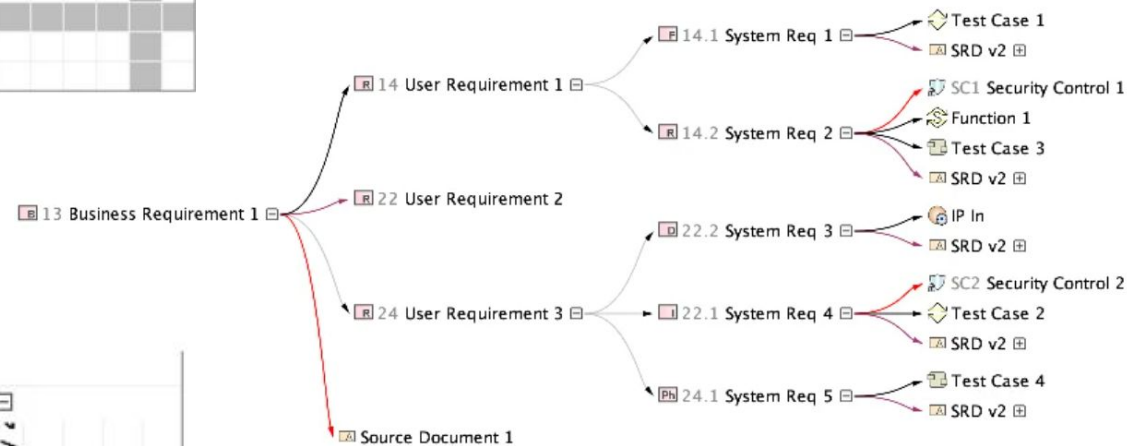
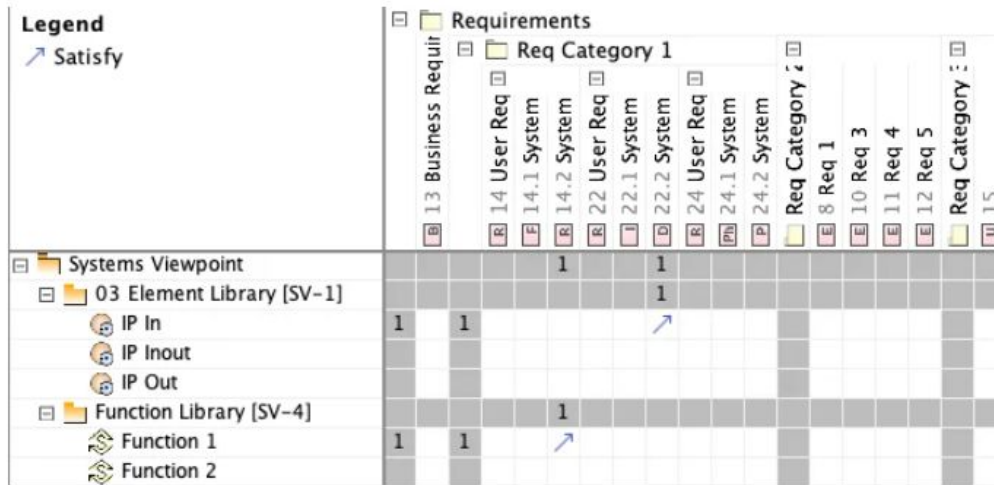
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Robust Requirements Management



MBSE provides the necessary ***traceability between capability, requirement, function and deployment*** – capturing details often lost to tribal knowledge when answering “why/what/when/who/how?” regarding implementations of complex SoS designs. These indirect benefits are not reflected in the labor hour metrics, but provide additional benefits when applied to a large SoS (Rogers III and Mitchell, 2021).

MBSE and Requirements Management



Examples of dependency matrices (Shevchenko, 2021)



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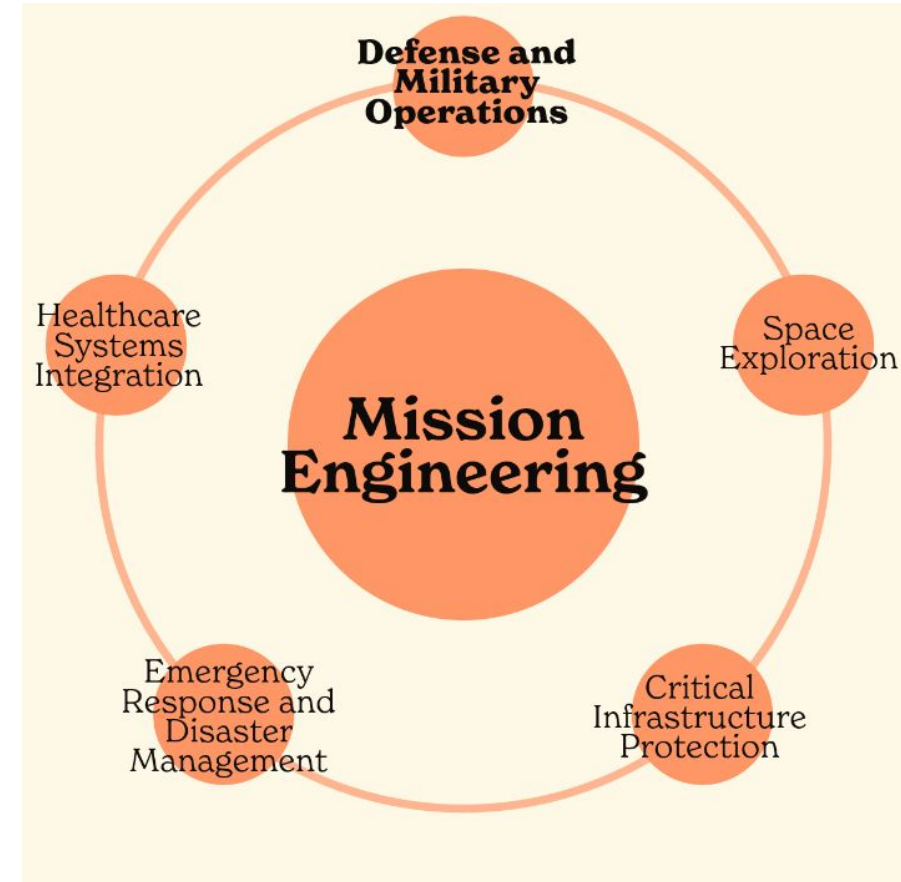
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Mission Engineering (ME)

The term “**mission engineering**” refers to the mission as the system of interest and applies systems engineering concepts, methods, and tools to the design of missions.

ME involves the determination of mission capabilities, a concept of operations, development and design of the operational activities, the operational network, and the allocation of operational activities to operational nodes for execution. In many systems, the operational nodes are different systems, thus leading to **a mission-oriented system-of-systems (SoS)**, which is formed to execute a single mission after which the SoS would disband.

In the military, the term mission defines the objective of a military operation. However, the concept of mission engineering could be more broadly applied in other organizational contexts from logistics, healthcare, to energy since we design all systems with a purpose or objective.

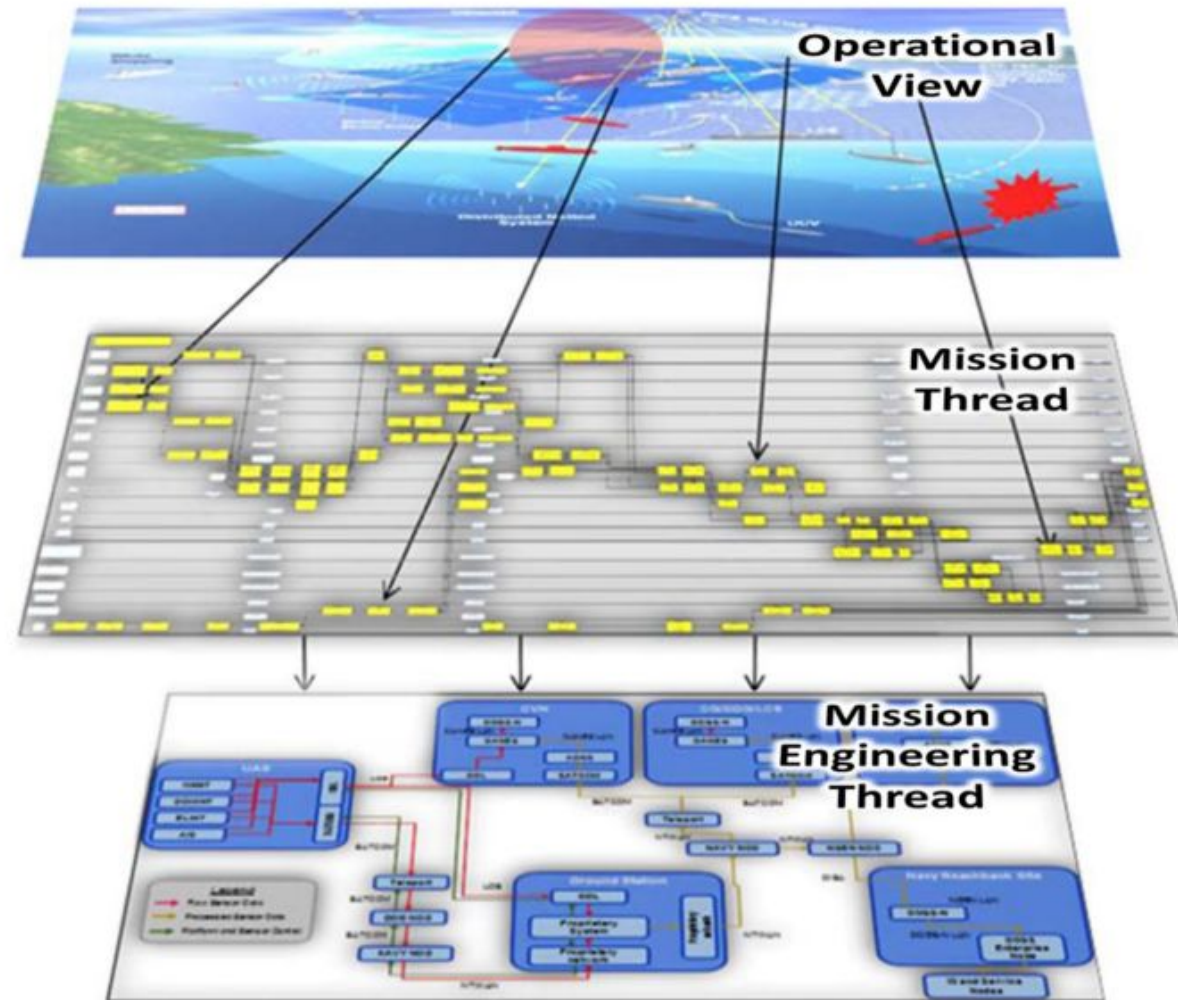


Mission Engineering (ME)

In ME, there are two key elements of mission architectures:

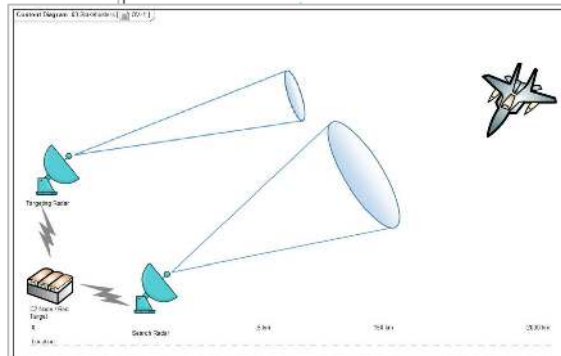
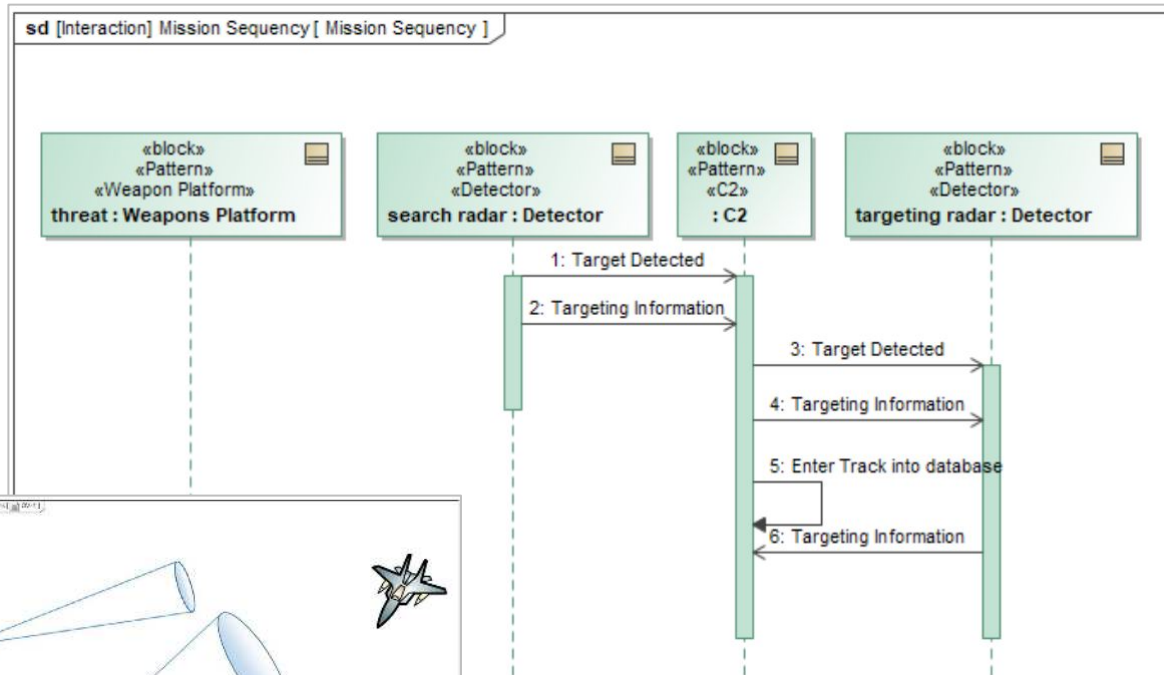
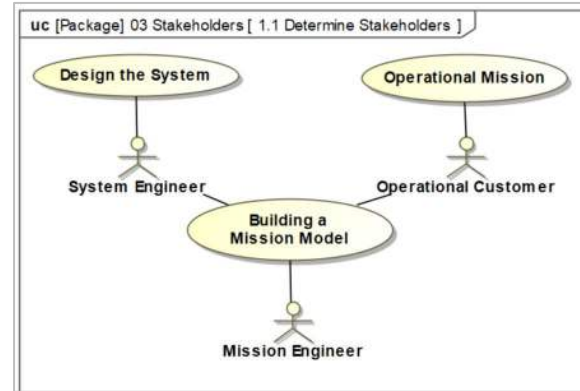
- 1) **mission threads**, which capture the activities of a given mission approach, and
- 2) **Mission Engineering Threads (METs)**. These elements capture how the mission activities related to the actors, systems, and organizations are executed in a specific mission context captured in the scenario and related vignettes.

There are multiple ways to document a mission architecture and several notational approaches that can be used to describe mission threads and METs, including SysML.

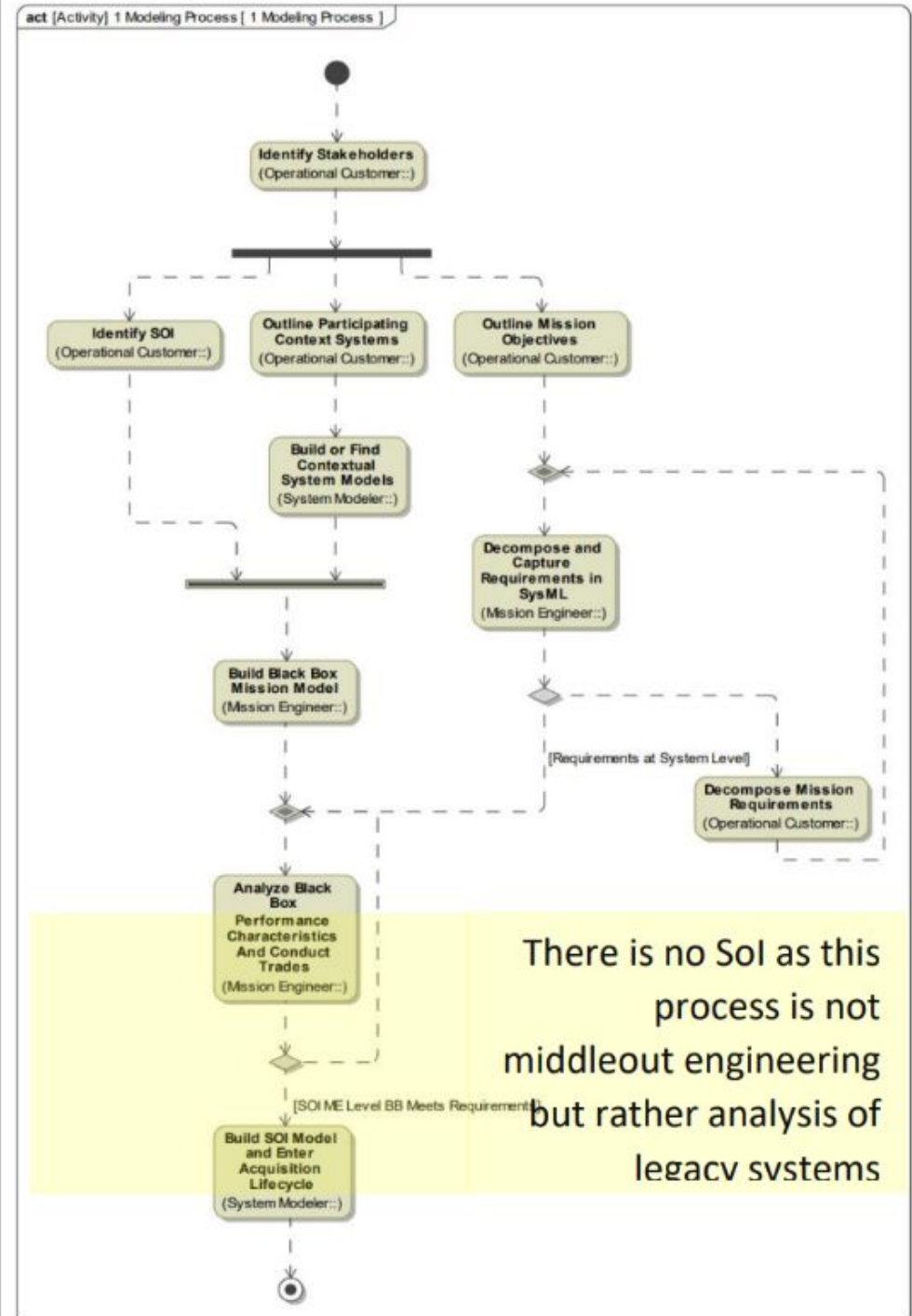


DoD (2023)

MBSE and ME



Example of MBSE Methods Applied to the Mission Engineering Modeling Use Case (Kidner, 2022)

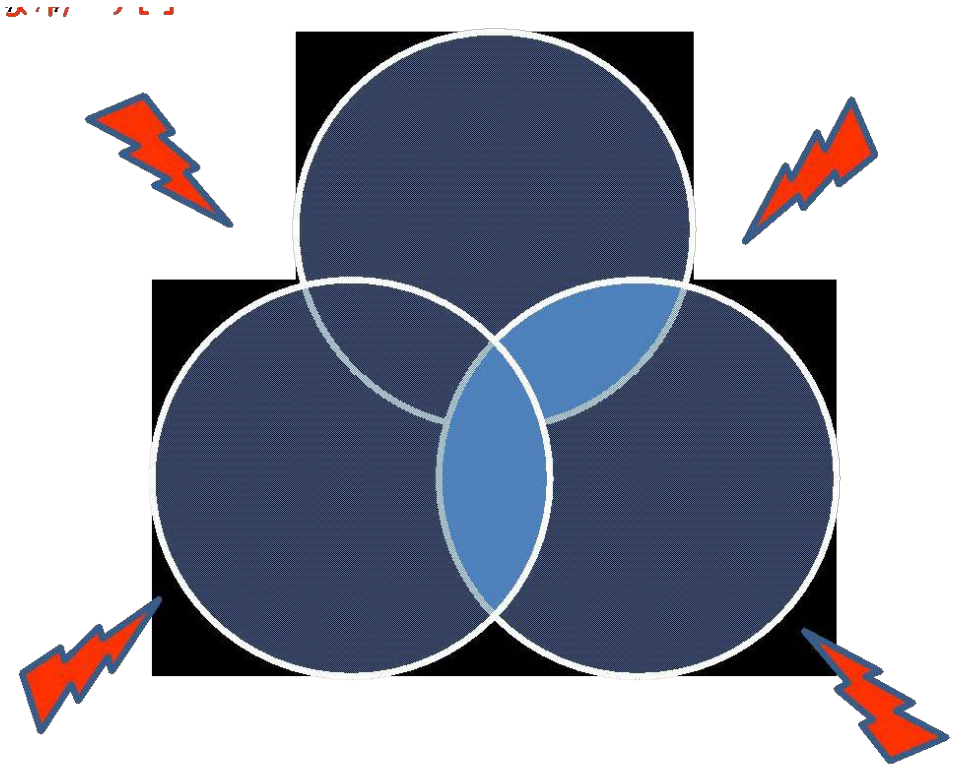


There is no SOI as this process is not middleout engineering but rather analysis of legacy systems

Criticality of Cybersecurity Assessments

MBSE helps deal with the complexity of advanced persistent threats (APTs) when attacking complicated cyber-physical systems like weapon systems and critical infrastructure. It aids in traceability, risk-based informed decision-making, agility and through-life management.

The complexity (emergence) in cybersecurity comes predominantly from evolving cyber threats, especially advanced persistent threats



Cyber Threat Categories and Vulnerable Systems (GAO, 2018)

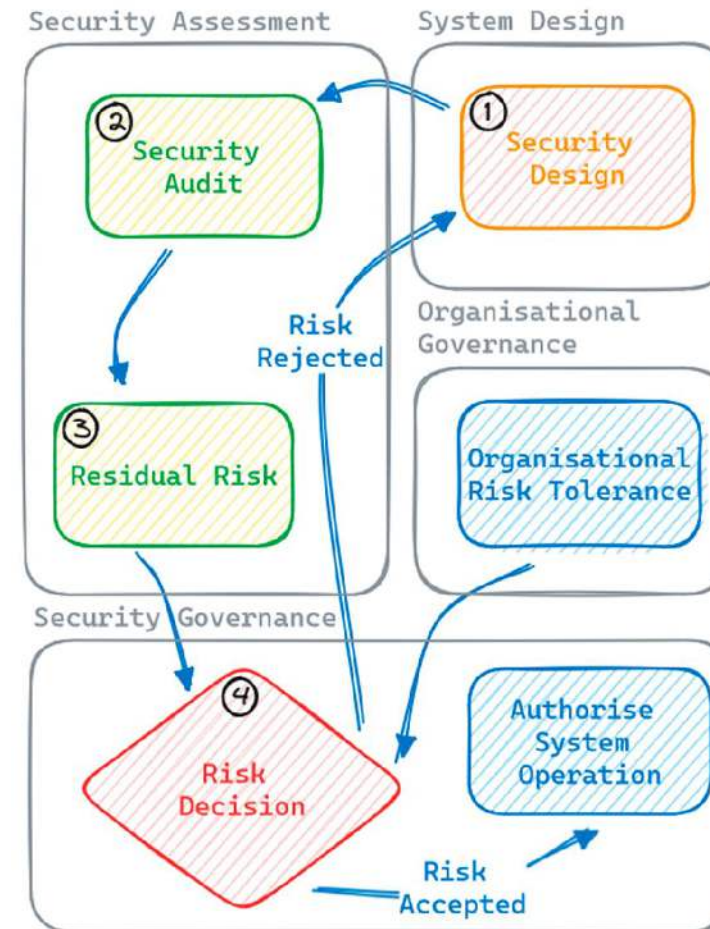
Criticality of Cybersecurity Assessments

Model-based assurance (MBA) is a process that supports system verification and validation requirements using conceptual and analytical modeling and simulation techniques.

With the increasing prevalence of digital engineering and model-based systems engineering (MBSE), there is potential to transform traditional system assurance processes to more holistic and evidence-based forms using models.

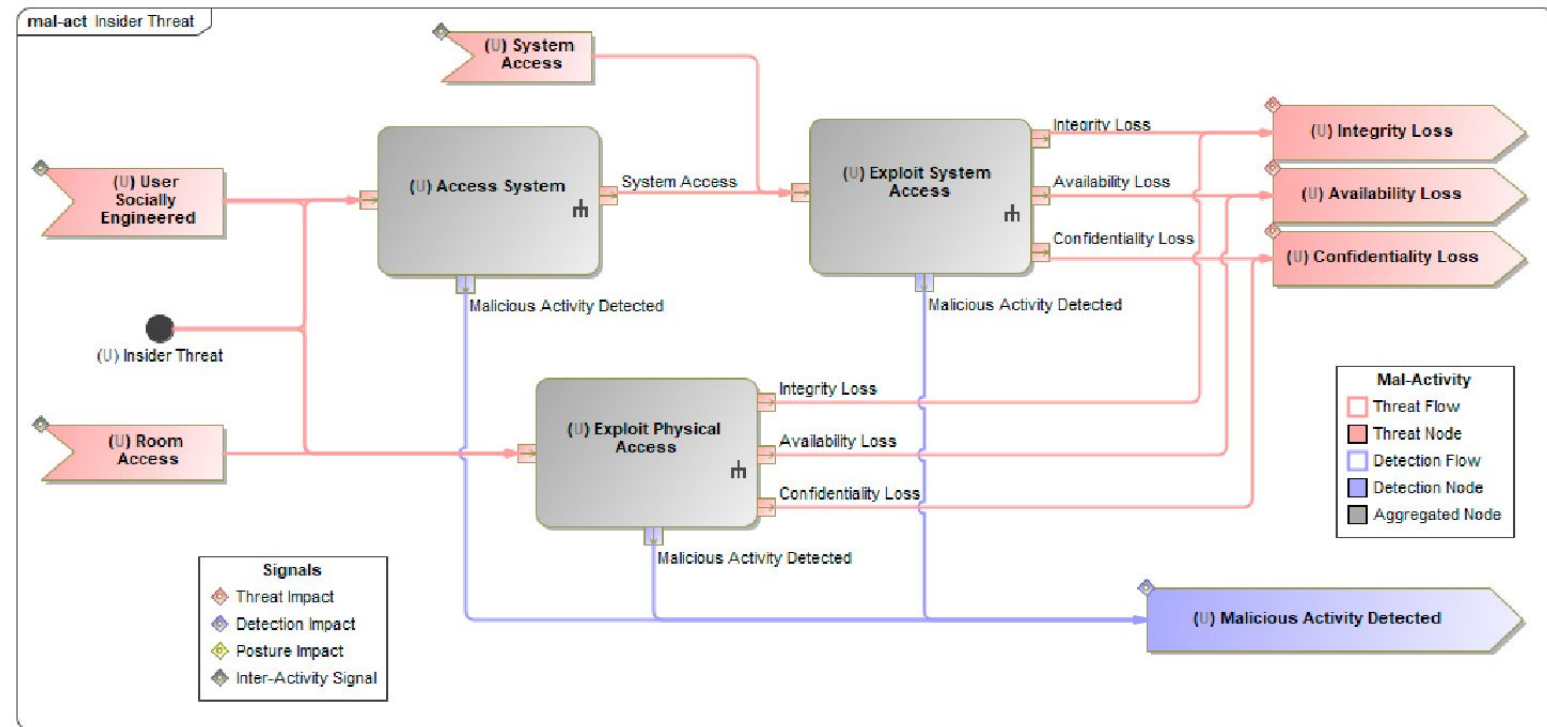
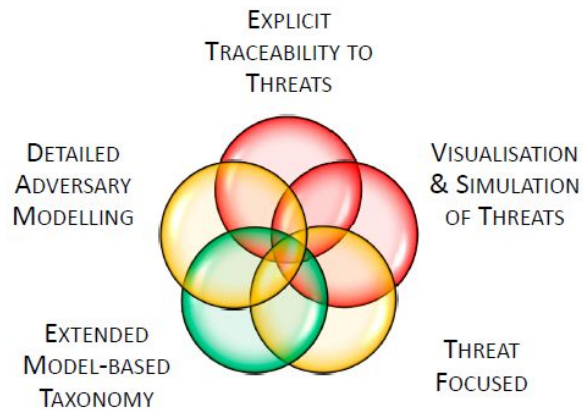
In most organisations mission assurance and safety are the responsibility of traditional engineering (technical regulatory frameworks), whereas security and cybersecurity are usually separate responsibilities (i.e., information assurance), with the latter being substantially more enduring due to threat evolution.

Engaging and connecting both security and engineering in common model-based assurance and then testing is very challenging.



Generic process for making cybersecurity risk decisions under a compliance-driven risk assessment paradigm (Fowler et al., 2024)

MBSE and Cybersecurity Assessments

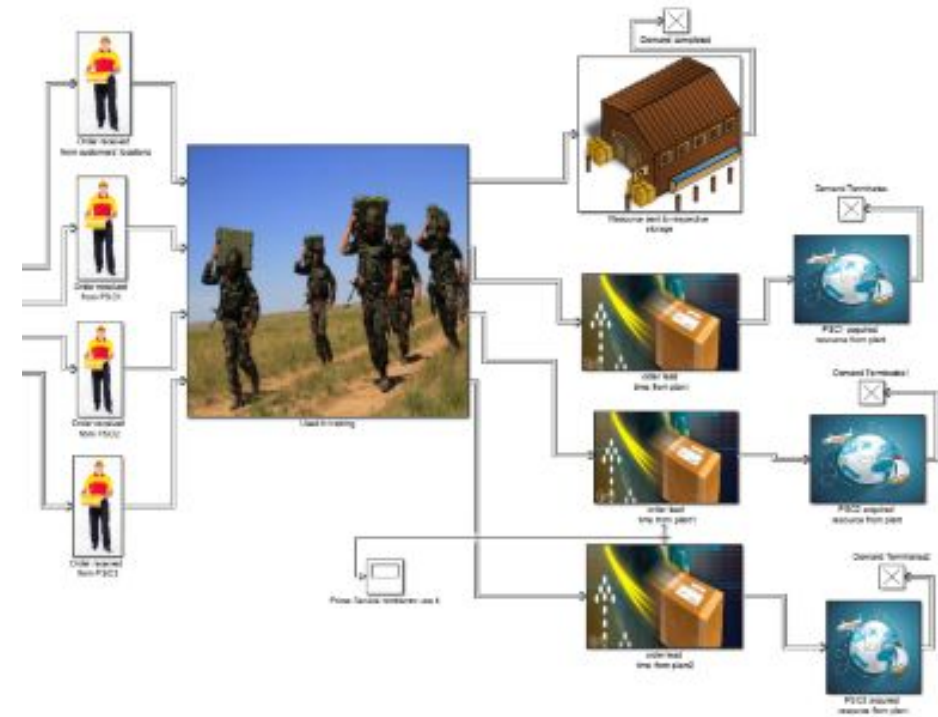


Intermediate CEMT mal-activity diagram for the Insider Threat misuse case (Fowler et al., 2024)

Key to achieving the features that inform decision-making are mal-activity diagrams showing the kill-chains and key controls, which then sum to risk assessments.

Supply Chain Management

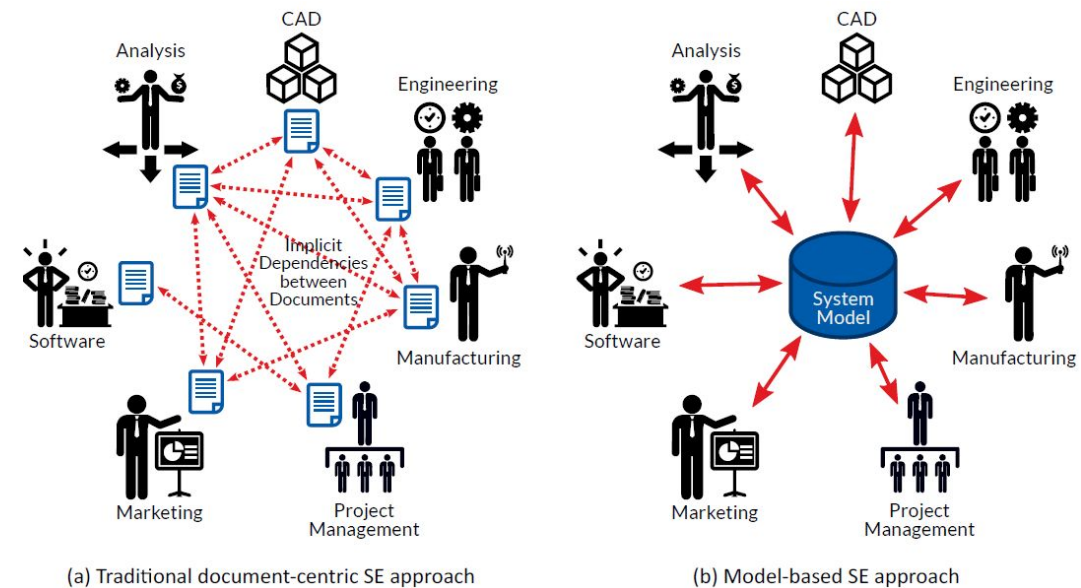
MBSE helps deal with the complexity caused by globalisation, conflict and climate responses. It aids in traceability, risk-based informed decision-making, agility and through-life management.



“The use of models in engineering is not a new phenomenon; however, the use of models in systems engineering to replace the previous document-centric approach (as mentioned above) is increasing in usage. Yet, application of MBSE is **unbalanced across the system lifecycle**. Usage is mostly limited to requirements and functional definition and decomposition. With the many methodologies that have been developed to utilize MBSE, investments in the expansion of MBSE to more stages of the development lifecycle continue. Even with established methodologies growing, specific application of MBSE to obsolescence management remains lacking.” (Morgan et al. 2021).

MBSE and SCM

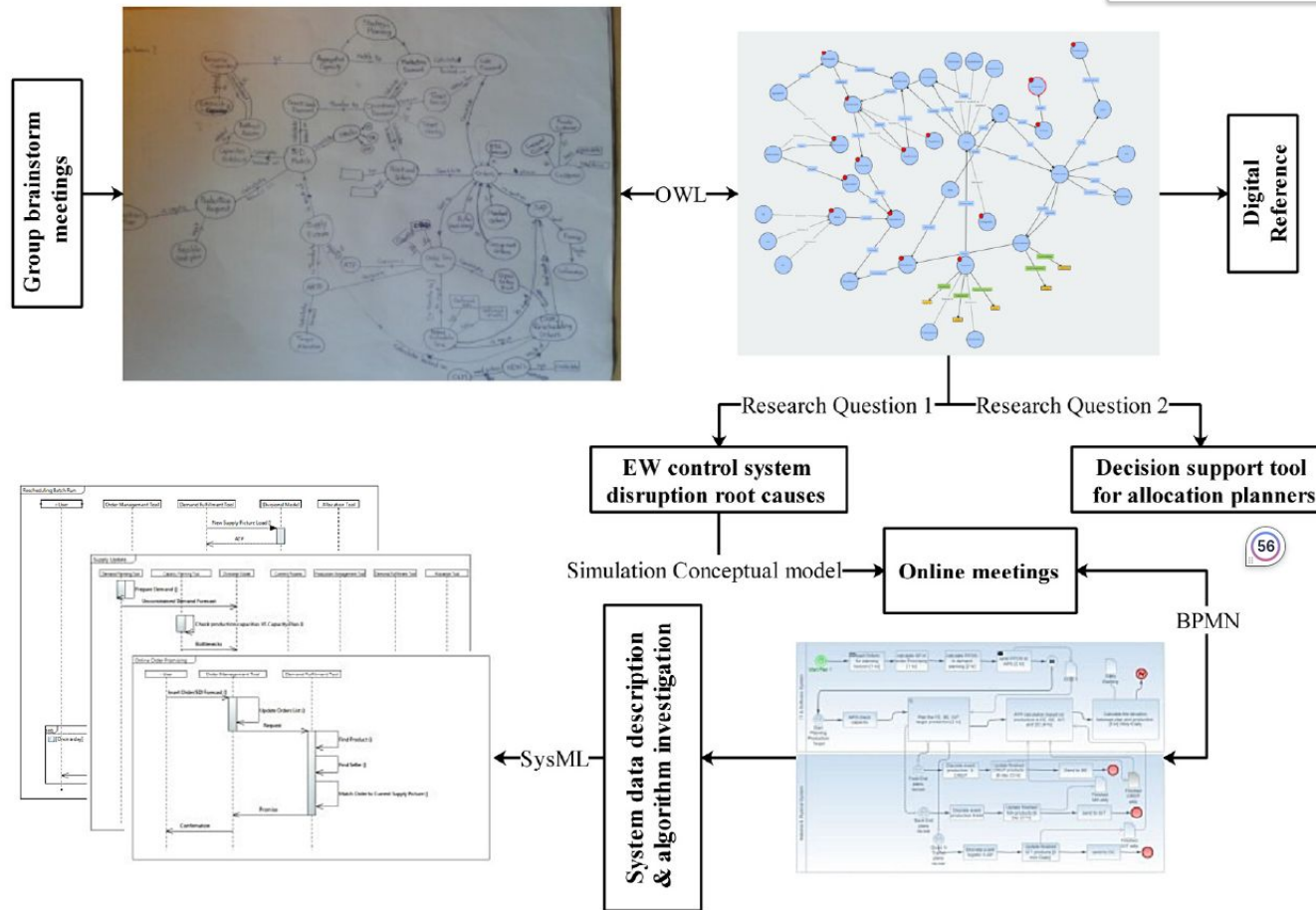
MBSE is noted to achieve the following key SCM benefits:



- **Reduce errors.** Reduce error rate, earlier error detection, reduction of failure corrections, limit human errors, early detection of issues, detect defects earlier, early detection of errors and omissions, reduced specification defects, reduce defects, remove human sources of errors, reduce requirements defects.
- **Reduce time.** Shorter design cycles, time savings, faster time to market, ability to meet schedule, reduce development time, time to search for information reduced, reduce product cycle time, delays reduced.
- **Higher level of support for integration.** Integration of information, providing a foundation to integrate diverse models, system design integration, support for virtual enterprise/supply chain integration, integration as you go.

The advantages of using an MBSE approach to SCM is leveraging the modelling to do a concurrent approach where decisions about supply chain and manufacturing are addressed whilst the design configuration is decided.

MBSE and SCM



Pathway for implementing and using MBSE approaches for system understanding, communication with stakeholder, define research question, conceptual modeling, and depth module investigation (Mousavi et al., 2022)

Part III: MBSE Adoption and Integration



MBSE Adoption: Challenges

- Achieving harmony between the MBSE elements.
- Purpose driven MBSE methodologies.
- Awareness & change resistance.
- Executive level sponsorship.
- Modularity & reusability.
- Large models visualization & communication.



MBSE Adoption: Strategies – MBSE as an Improvement Process

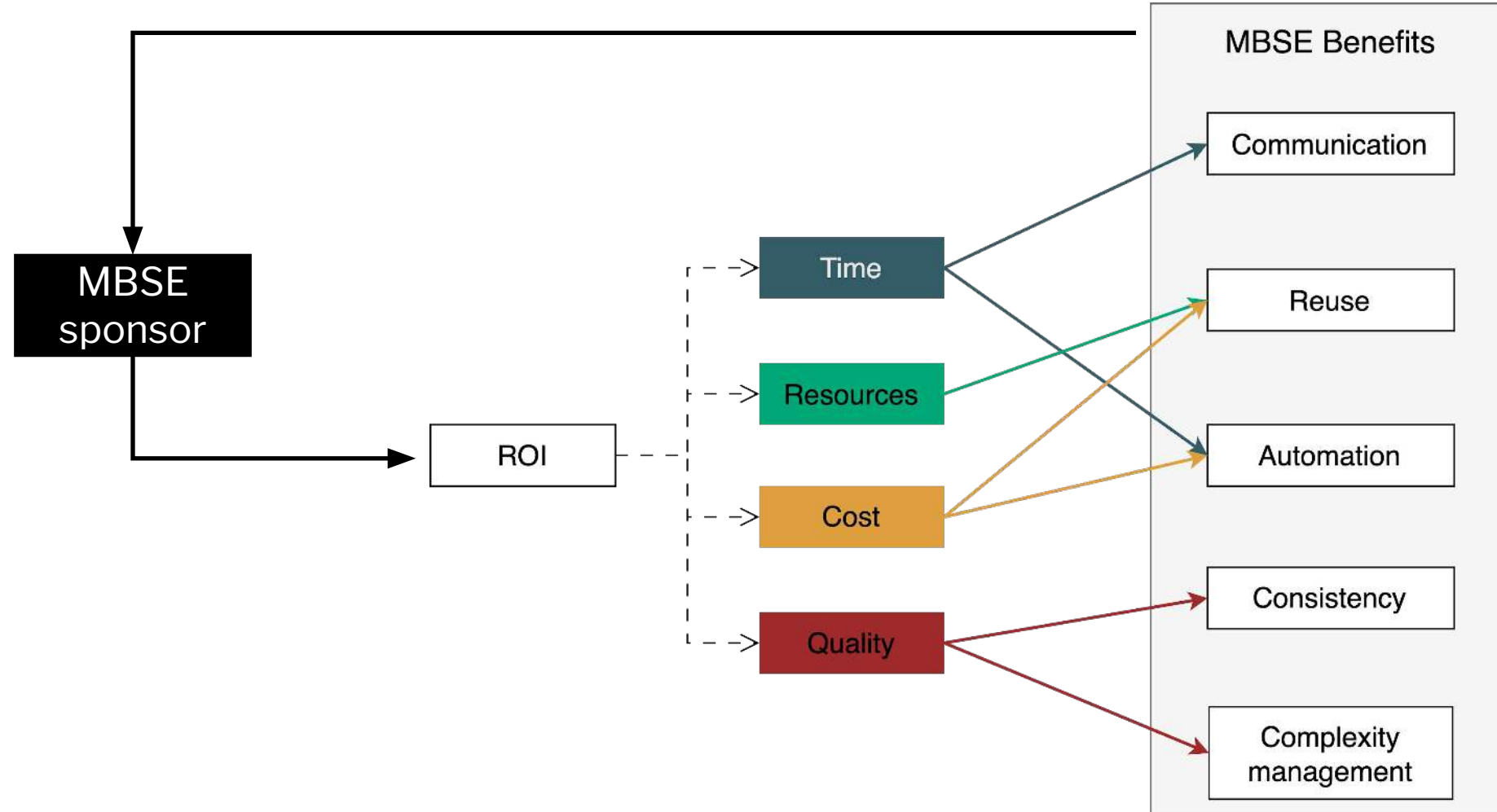
1. Monitor & asses.
2. Plan the improvement.
3. Define the changes.
 - Processes.
 - Methods.
 - Tools.
 - Trainings.
4. Pilot the approach.
5. Incremental deployment.



MBSE Adoption: Strategies – Trinity Approach

- Stakeholder mapping
 - User, operator, system sponsor, manager, engineer,...
- Stakeholder needs mapping
 - Engineer: improve system development, manage complexity, improve automation, improve tool interoperability.
 - MBSE sponsor: demonstrate ROI, increase quality, increase sales.
- Create benefits map
 - Map stakeholder needs to MBSE benefits, then communicate accordingly.

MBSE Adoption: Strategies



Part IV: Industry-Focused MBSE Curriculum



Capability Systems Centre (CSC), UNSW Canberra

The **Capability Systems Centre** is a world-leading research institute and trusted go-to for Government and industry in the critical area of capability management and all aspects related to the capability lifecycle (design, acquisition, sustainment).

We undertake high-quality research in key areas internationally recognised as being innovative and interdisciplinary, including complex systems science, computational modelling and informatics, decision science, test design, and actionable science.

Education and Training

Research and independent advice

Mentoring

Modelling and simulation research

Technical deep dives and investigations

Technical risk reduction

- Research on a range of topics associated with **force design** (mission engineering)
- Facilitated workshops



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CSC Industry-Focused MBSE Curriculum: Role-Specific Capabilities to Meet Industry Needs

L1

MBSE sponsor



Supports and promotes MBSE practices, aligning them with strategic goals for smooth organizational integration, e.g., CTO (Chief Technology officer), Systems Engineering Manager.

L1

L2

Operational Stakeholders



Individuals engaging with MBSE processes, offering feedback and information to shape MBSE practices, e.g., Quality assurance engineer, Design engineer, Operations manager.

L1

L2

External partners



Collaborators and clients who use MBSE deliverables for purposes of compliance, requirements alignment, and logistics, e.g., Client project manager, compliance officer, supply chain analyst.

L1

L2

L3

L4

MBSE Core Team



The team of specialists dedicated to guiding the organization's MBSE efforts, e.g., System Architect, MBSE tool specialist, Systems Modeler.

MBSE competencies

This level delves into analyzing, building, and maintaining accurate and consistent models, both individually and collaboratively. It emphasizes integrating diverse engineering models with the system model, as well as developing models for complex systems and systems of systems (SOS).

This level covers comprehensive analysis of system models, collaborative model building and maintenance for accuracy, integration of diverse knowledge domains, adherence to development standards, and model utilization for engineering decisions throughout the system's lifecycle.

Upon completing this level, learners will develop an understanding of model artifacts, and their roles in enhancing interoperability and modularity. They'll learn to apply MBSE methods and use its tools for system modeling, utilizing models for design analysis and decision support. Additionally, participants will leverage model artifacts for effective system representation and stakeholder communication.

Upon completing this level, learners will grasp the fundamentals of MBSE and its advantages over traditional methods, understand the MBSE lifecycle and model applications, and learn how to evaluate the quality of models, methods, and tools. They will also acquire insights into workforce requirements for MBSE, emphasizing the role of model artifacts throughout a system's lifecycle.

MBSE course content

- Techniques for integrating MBSE with other engineering disciplines to enhance system architecture, focusing on complex systems and system of systems (SOS).
- Utilize MBSE model simulation for design validation and performance analysis.
- Model verification and validation.
- Identifying and correcting errors in complex systems models.
- Model interoperability and reusability.
- MBSE patterns and model libraries.
- Modeling data and data structures.
- MBSE as part of the digital enterprise - advanced strategies and Methods.

- Deep dive into analyzing and building complex system models.
- Techniques for effective team-based modeling and keeping model artifacts consistent and up-to-date.
- Strategies for integrating disparate model domains into a cohesive system model.
- Model maintenance and integration with industry standards.
- Utilize the system to inform decision-making and report generation.
- Requirements integration and traceability.

- Basics of MBSE modeling and common modeling pitfalls.
- Language-based modeling.
- Building MBSE models utilizing a language, a method, and a tool.
- Interpreting models and producing stakeholder-specific model views.
- Develop model artifacts according to intent.
- Use system model for design analysis.

- Introduction to MBSE and overview of its significance to systems' development.
- Economic analysis of MBSE processes.
- Basic concepts and terminology of MBSE (Languages, Methods, Tools).
- Quality of MBSE elements - Quality assessment of methods, tools, and models.
- Understanding model artifacts and their usage across the system lifecycle.
- Workforce development in MBSE contexts.
- MBSE adoption - challenges and strategies.
- MBSE as part of the digital enterprise - an Introduction.

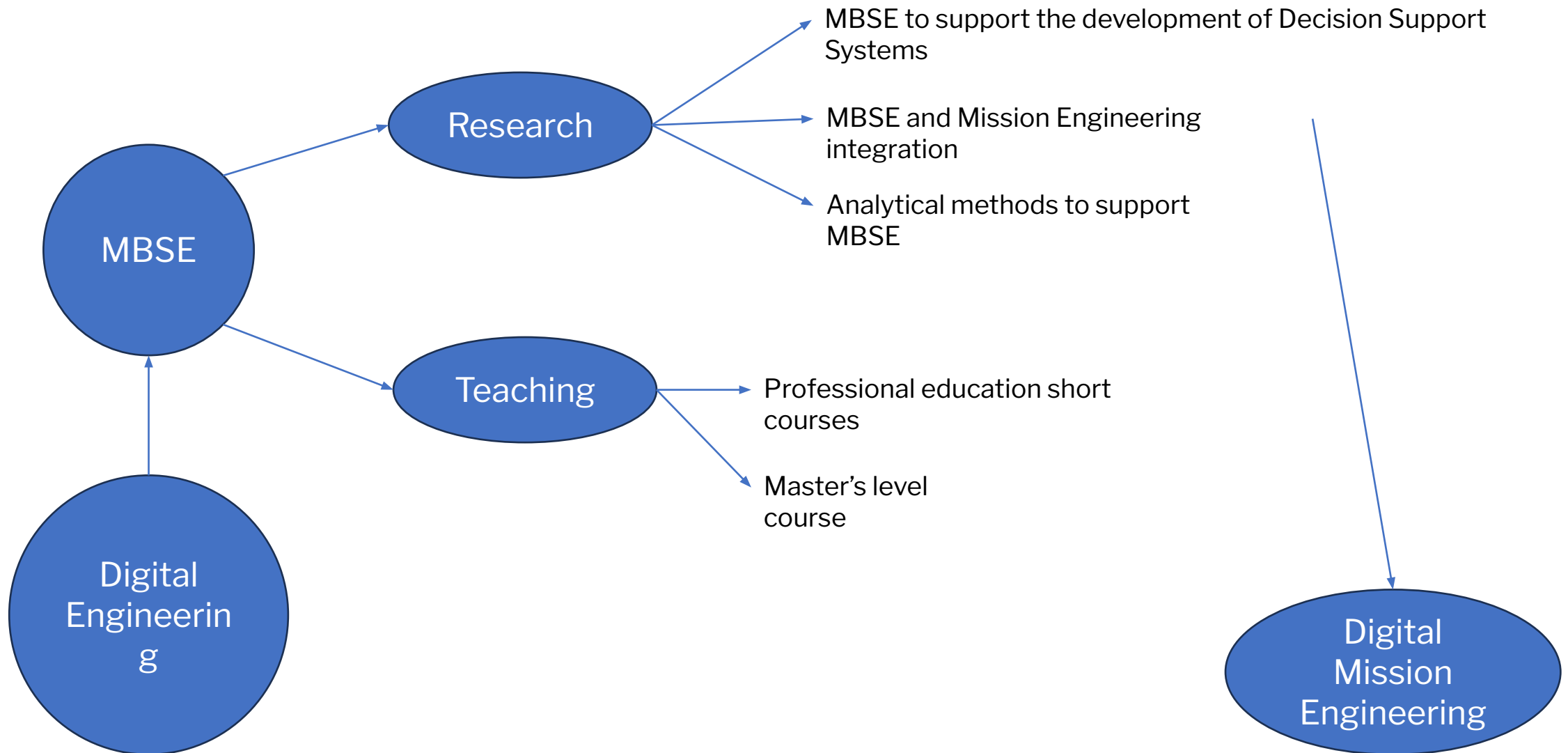


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CSC MBSE and Mission Engineering Scene



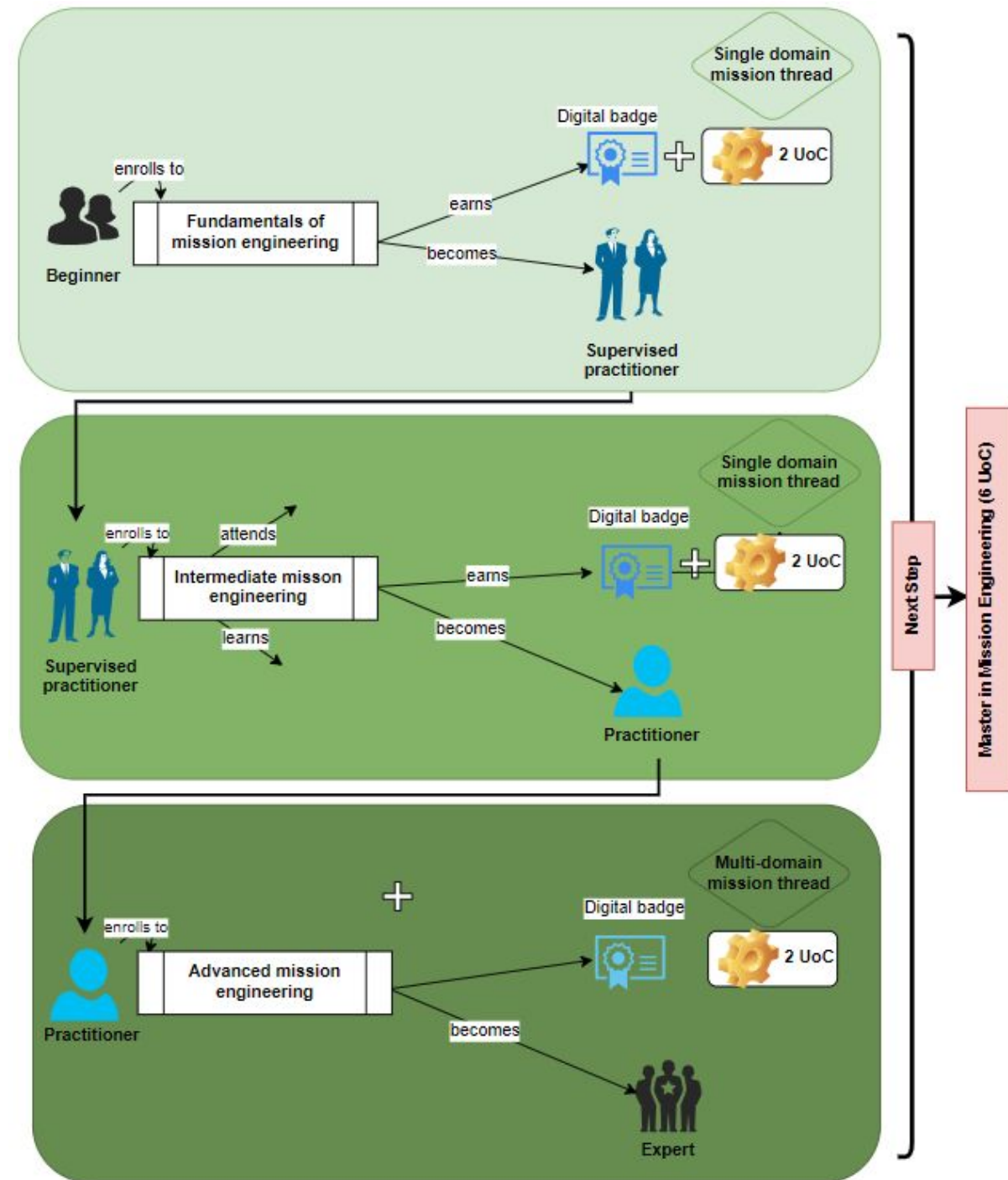
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CSC Mission Engineering Micro-credentials Curriculum

- A 3-tier micro-credential educational model is planned. (Fundamentals, Intermediate and Advanced).
- Key features
 - Skill transfer (ME competency framework)
 - Co-designed with industry
 - Accommodates beyond defence
 - Flexible learning journey for the learners



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The CSC Team

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